



Evaluating a Reinforcement Learning Approach for Collision Avoidance with Heterogeneous Aircraft

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This paper focuses on the problem of decentralized collision avoidance for Unmanned Aircraft Systems. The considered setting involves drones that can sense the presence of other traffic within a specified sensing area, for example due to remote ID signal broadcasts or visual perception, and take evasive maneuvers according to drone dynamics to ensure a minimum separation. A Reinforcement Learning approach is investigated to adapt the ego drone trajectory in response to the limited observations of the intruders trajectory within a bounded environment. The key aim of the work is on designing evasive maneuvers for ego drone when sharing the airspace with heterogeneous aircraft that have varying sensing capabilities, maneuverability, and risk-awareness. Beside reachability-based techniques, which offer a powerful framework for identifying safe trajectories under worst case actions by other agents, the proposed approach aims at adapting the evasive maneuver to the incoming aircraft behavior. Tests were executed in the simulated environment comparing the results obtained with heterogeneous incoming aircraft with different maneuverability levels and with randomly selected fixed obstacles. The Reinforcement Learning based method performance was also tested adopting different state vector parameters. The results show that the proposed approach can support the implementation of safe collision avoidance services allowing the generation of adaptive maneuvers for Unmanned Aerial System Traffic Management.

I. Nomenclature

$D1$	=	Ego drone
$D2$	=	Incoming drone
A	=	Action set
a	=	Action that belongs to the action set, i.e. turning maneuver of ego drone
n_{SMI}	=	Number of Separation Minima Infringement occurrences
N_{tot}	=	Total number of considered episodes
R	=	Reward
S	=	State space
s	=	State that belongs to the state space
s'	=	State at next time step that belongs to the state space
T	=	Transition function

II. Introduction

THE increasing market demand of Unmanned Aerial Systems (UASs), commonly named drones, is driving the development of advanced techniques for a safe integration in the civil airspace. Specifically, the presence of heterogeneous air vehicles in civil airspace has a strong impact on airspace regulation, technology development, and

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operative services. The consequence of heterogeneous vehicles acting in the same volume is that the dynamics of all agents can not be well known or cannot be easily inferred by the ego drone due to limited observability. Drones with different specifications need dedicated air services. Package delivery, search and rescue, air taxiing, environmental monitoring applications involve drones with different characteristics and performance that must be managed guaranteeing a proper safety level. Innovative solutions and services are worldwide required to support the integration of unmanned platforms in the civil airspace in the context of UAS Traffic Management (UTM), and a great effort is needed by the aviation authorities to properly manage this advanced paradigm, such as [1], [2], and specifically for Urban Air Mobility (UAM), as reported in [3], [4], [5], [6].

A. Traffic Management Services

To allow safe operations of UASs, several services must be defined to manage the expected demand with an increased level of automation. Different layers can be designed to provide adequate support to UAS operator and Air Traffic Control (ATC) during all mission phases. At strategical phase, routes proposed by UAS operators must be approved or suggestions can be provided to guarantee free collision paths. Path planning algorithms help the definition of safe routes in a multi-agents scenario. Proper identification and localization techniques must be provided to recognize specific drones for surveillance purpose or in case of emergency conditions. The definition of the on-board sensor suite must enable safe operations for collision avoidance and tracking purposes. Cooperative and non-cooperative solutions must be implemented according to airspace rules and vehicles specifications to guarantee the same level of reliability and robustness in the multi-platform and multi-mission scenario. Ground-based infrastructures must be designed to support the high expected demand of unmanned platforms interacting with the ATC if needed. Table 1 reports some worldwide key programs, plans, and projects that support the definition of a systematic development of UAS operations.

Table 1 Programs, Plans and Projects for UAS traffic management.

Name	Main Features
UTM Pilot Program [7]	Development of services for UTM with a proof of concept for UTM capabilities
UTM Fiel Test [8]	Validation of more complex UAS operations for improving UTM
UTM Impementation Plan [9]	Implementations features for UTM services and operations
U-space Blueprint [10]	U-space vision and intents by SESAR Joint Undertaking
CORUS [11]	Concept of Operations for low altitude airspace

In addition, among the several services that must be implemented to assure safe operations, specific projects investigate the separation task, such as [12], [13], [14], and conflict detection and avoidance task, such as [15], [16].

B. Related Works

Planning task is needed to identify safe routes of multiple agents according to proper requirements and thanks to advanced techniques, such as exploiting Sequential Trajectory Planning algorithms [17], or by using specific maps [18], [19]. Collision avoidance task must be implemented developing standard procedures to manage risk situations, as reported in [20]. considering the UTM/UAM scenario, Reinforcement Learning (RL) based techniques can support demand balance [21], [22], optimal flight policies [23], path planning [24], conflict resolution [25], [26]. The use of meta-RL approaches can also develop advanced conflict resolution strategies [27]. The adoption of a continuous maneuver range can support RL based conflict resolution [28].

In the context of collision avoidance services, this paper describes an adaptive implementation of a RL based algorithm to generate evasive maneuvers. Besides reachability based techniques that usually assume worst case behavior by all agents, generating extremely conservative maneuvers, the key challenge is in identifying safe evasive trajectories that are not extremely conservative by being cognizant of semi-cooperative response by heterogeneous agents with unknown preferences, avoiding that a collision between agents leads to an unfavorable outcome for all agents. This

paper involves the following sections: Section III describes the simulation environment and the characteristics of the considered drones and algorithm, Section IV includes the details about the developed simulations, Section V discusses the achieved results by comparing the proposed approach with benchmarks, and the Section VI presents the conclusion of the conducted activity.

III. Methodology

For simplicity, the considered environment includes 2D trajectories and it is populated by one ego drone, fixed obstacles, and two types of incoming aircraft: a multi-rotor drone and a fixed-wing drone. Each type of incoming aircraft has two levels of maneuverability. We use the Soft Actor Critic (SAC) algorithm to train a RL agent to learn maneuvers for the ego drone for scenarios having homogeneous as well as heterogeneous drones. The Separation Minima Infringement (SMI) is considered as a condition where the selected criteria to define a safe separation are not satisfied. In the proposed approach the frequency of SMI, i.e. f_{SMI} , is computed as the percentage value according to Eq.1, where n_{SMI} is the number of SMI occurrences and N_{tot} is the total number of considered episodes.

$$f_{SMI} = \frac{n_{SMI}}{N_{tot}} \quad (1)$$

The developed methodology is summarized in Fig.1. The environment set includes the definition of the operative area and the drone behaviour. The operative area is the airspace sector where the operations will be executed. Proper thresholds must be considered to define the arrival point area and the collision scenario. Bounded areas must also be defined to limit the airspace where drones can apply evasive maneuvers in order to reduce the risk of the so-called domino effect. The airspace features defined by the mentioned parameters affect the drone behaviour. However, besides the external factors related to the operative environment, additional specifications related to the drone trajectory and constraints must be added, such as velocity limits and turning angles. In this paper, the scenario involves two drones, but many vehicles can be added in the proposed method. The environment definition allows to proper definite the input parameters to the collision avoidance model. After the selection of the most suitable algorithm, the characteristic episode scenario that allows to train the model must be fixed. The environment features also allow to define the state vector and the required output that support the generation of the desired evasive maneuver. According to a RL approach, the reward values must be set and a benchmark must be described to assess the performance of the proposed method.

This paper investigates a RL based methodology for collision avoidance in the UTM scenario. However, the proper identification of operative area characteristics and the vehicle behavior allow to generalize this method also for other terrestrial and maritime environments populated by unmanned vehicles.

A. Environment Definition

The environment is set considering two drones, i.e. the ego drone D1 and the incoming drone D2. Each drone position vector $[x, y]$ is limited in a bounded airspace that represents the area where the ego drone must maneuver in case of collision where $x \in [-400m, 400m]$ and $y \in [-400m, 400m]$. In this paper, a two-agents scenario is investigated and visualized, thanks to the library [29]. The characteristics of involved drones are preliminary selected:

- Ego drone D1. The ego drone initial position is located at the center of the area at coordinates $[0m, 0m]$ and the destination position is randomly selected in the bounded area in each episode. The drone is supposed as a multi-rotor drone with a constant velocity of 20m/s. In the bounded area the ego drone executes a straight path trajectory to reach the planned destination if no collision occurs. Otherwise, for each time step it can execute a turning action in a continuous range $[-90^\circ, +90^\circ]$ according to the adopted RL algorithm described in Section IIIC.
- Incoming drone D2. Each incoming drone is initialized according to the class it belongs to. The initial and final positions are randomly selected in the bounded area considering a threshold of at least 48m between the initial position of the ego vector and the initial position of the incoming drone. The velocity module is respectively 8m/s or 32m/s for multi-rotor and fixed-wing drones. A course angle change of 60° is set for multi-rotor drones and a left or right turn with a rate of $1^\circ/s$ is set for fixed-wing drones. Each drone should reach the planned target destination by executing a straight path. Turning maneuvers are executed for collision avoidance purposes. When the intruder senses that the relative distance is smaller than the assigned threshold, it starts an evasive maneuver. The incoming drone trajectory is stopped when the ego drone reaches its destination point.

The relative distance between the ego drone and the incoming drone is computed at each time step to evaluate if a violation of minimum separation occurs. A threshold of 24m is set to identify the SMI condition. A maximum number of 70 time steps is also set to end the episode.

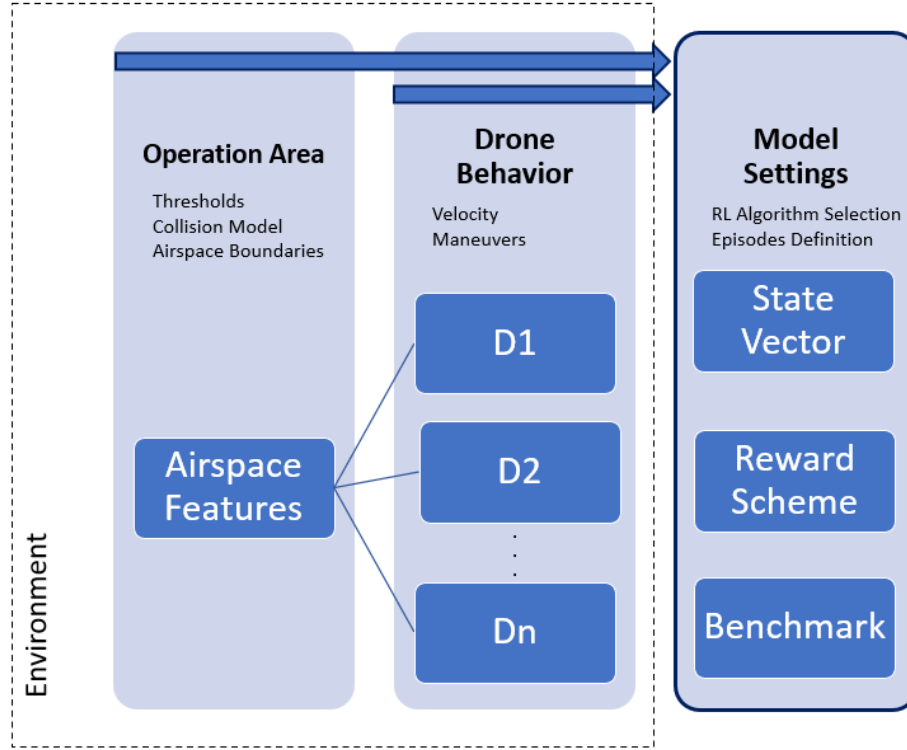


Fig. 1 Schematic diagram of the applied methodology.

An episode stops when the ego drone reaches the target destination, i.e. the planned final waypoint within an assigned threshold, or when the maximum number of time steps is reached. If the time of the maneuver is less than the maximum time, the drone will restore the direction to the destination point compensating for the direction change.

Fig.2 shows the described environment when a collision occurs, highlighting the ego drone trajectory in red and the incoming drone trajectory in blue in the defined maneuver area with transparency effects to show the collision point. Within the sensing area that is defined by the on-board systems configuration, size and shape of the maneuver area can be defined according to geographical limits, mission operations, geo-fence. In the proposed approach a fixed maneuver area is considered. The green threshold represents the destination area that must be reached to complete the drone mission, neglecting approach and landing phases. The orange circle identifies the violation of the collision area. In the proposed example, the simulation was not stopped after the collision threshold violation.

B. Problem Assessment

The need of an advanced method to generate an evasive maneuver in agreement with the incoming drone behavior is described considering two test case examples, i.e. Test A and Test B. The features that fix Test A and Test B characteristics are only related to different drones relative positions. To describe the problem assessment, the tests are focused on two multi-rotor drones, i.e. the ego drone and the incoming drone and no obstacles. Both tests involves straight path conditions that describe a collision event where both drone do not perform any evasive maneuvers and the simulation is stopped when the relative distance is lower than the threshold, as shown in Fig.3.

Several collision avoidance approaches consider the observation state vector at the previous time step that is used to generate the maneuvers to maximize the relative distance. Reachability based sets can be defined according to specific constrains and the benchmark proposed in this activity involves a turning angle constraint with a discrete step range of 20 actions in the range $[-90^\circ, +90^\circ]$. The drone behaviour is set according to Section IIIA. The obtained trajectories in Test A and Test B are respectively described in Fig. 4.

According to the proposed operative environment, the single step benchmark approach achieves full success with the slow multi-rotor incoming drone, while the frequency of SMI occurrences is $f_{SMI} = 6.66\%$ for the faster fixed-wing

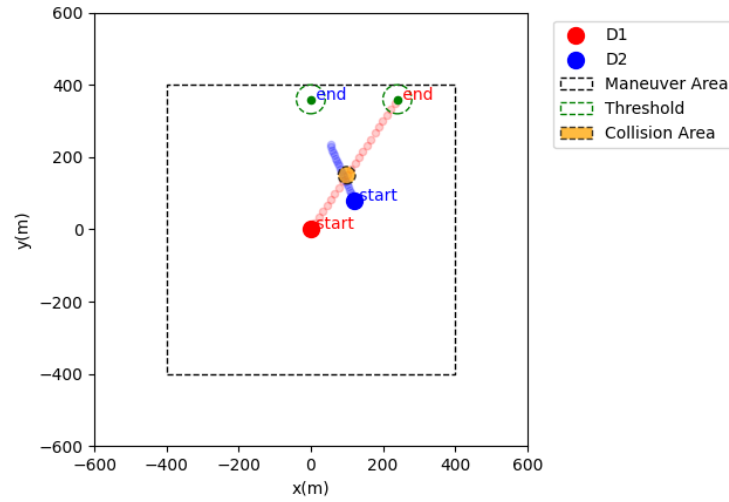


Fig. 2 Example of scenario description with ego drone and one incoming drone in the defined environment. The simulation is not stopped after the collision threshold violation.

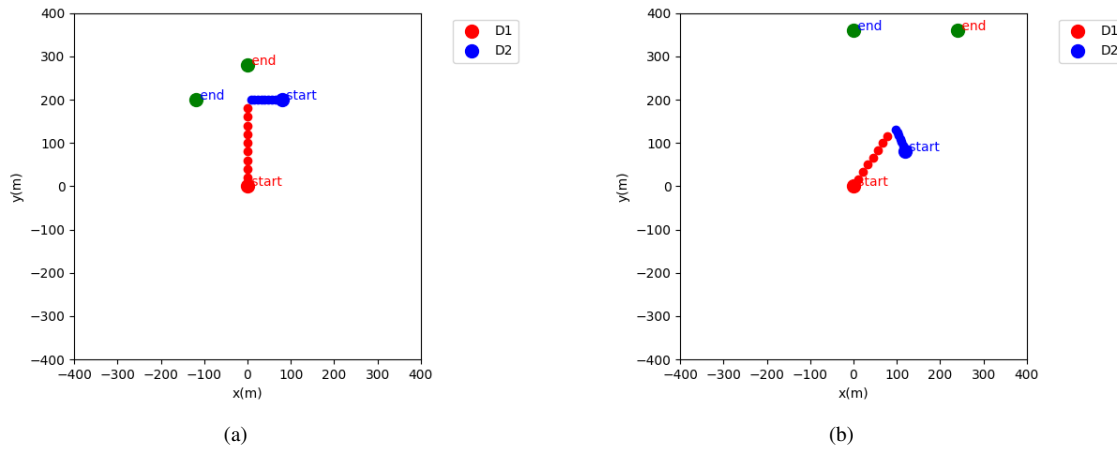


Fig. 3 Examples of straight path collisions in the maneuver area with a multi-rotor incoming drone. a) Test A scenario. b) Test B scenario.

drone due to the higher velocity drone that is more difficult to avoid considering only single time step approach.

Analyzing the obtained results, three main issues can be observed:

- The deterministic application of this approach involves the evaluation of all possible actions in the selected range and the computational time can not be feasible for a real-time application.
- The principle of evasive maneuver generation is based on maximizing the relative distance according to a high conservative approach.
- This approach is strongly sensible to the selected threshold. With the considered threshold of 24m is not possible avoiding all collision events in any cases.

A deterministic strategy that is adapted to the kinematics characteristics of the incoming drone can support the generation of effective and safe maneuvers exploiting a RL based approach.

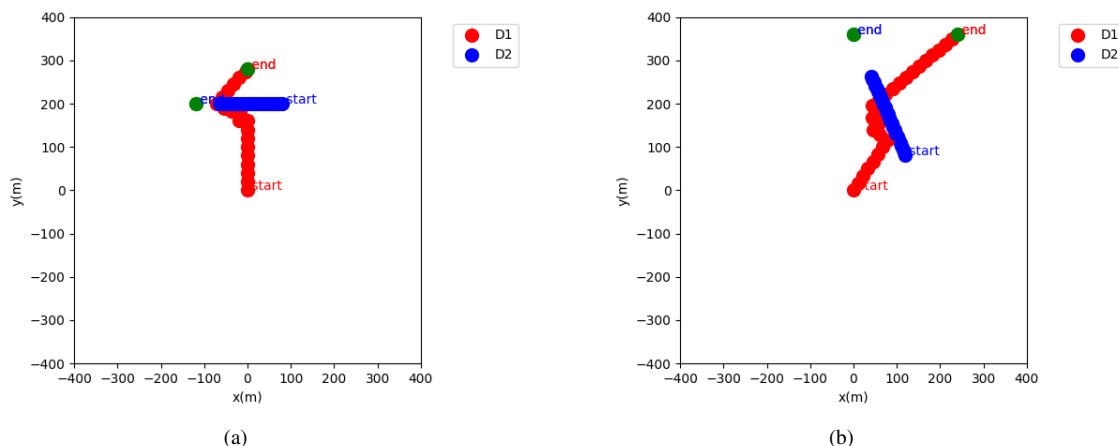


Fig. 4 Examples of evasive maneuvers based on maximizing the relative distance according to a discrete range of actions with a multi-rotor incoming drone. a) Test A scenario. b) Test B scenario.

C. Reinforcement Learning Algorithm

The Reinforcement Learning approach can be used to tune the performed evasive maneuver according to the incoming drone flight profile. A Markov Decision Process (MDP) can be considered, where S is the state space, A is the action set, and T is the transition function from the state $s \in S$ to the state $s' \in S$ given the action $a \in A$, i.e. $T(s, a, s') = Pr(s'|s, a)$. R is the reward defined as $R : S \times A \times S \rightarrow \mathbb{R}$. The action a should be selected according to the intruder flight profile to avoid the exploitation of the maximum actuator capability. If no prior knowledge about the models of T and R is available, a model-free solution of the MDP can be exploited adopting a RL based approach to maximize the expected reward. In the presented method, a Soft Actor Critic (SAC) algorithm is used.

The example reported in Fig.5 shows the trajectory of a multi-rotor drone from the initial point to the destination point avoiding a fixed obstacle.

Considering the defined environment, where an incoming drone is entering in the maneuver area, the drone must avoid a moving obstacle, without violating the delimited maneuver area. Also, the main goal of the proposed approach is tuning the avoidance maneuver to the incoming drone behaviour. To this purpose, the reward has been selected according to the following values:

- Occurring a SMI event. If a collision condition occurs a negative reward is considered $R = -2.5$.
 - Exiting the maneuver area. The ego drone must maneuver in a bounded area to avoid domino effect with drones in other airspace sectors and too large maneuvers. If the evasive maneuver is executed outside the bounded areas a negative reward is considered $R = -0.01$.
 - Reaching the destination point. If the drone reaches the destination point, a positive reward is considered $R = 1$.
- Considering the MDP, the next state s' is only related to s and a and not to historical data.

IV. Simulations Development

Several simulations were performed training the ego drone response to collision with different incoming drones with normalized input parameters. The SAC algorithm from stable-baseline3 project [30] was used considering a total number of 1500000 time steps for the training phase. the results are presented considering an approximation error of ± 0.05 over the percentage value. Three types of test were developed with an increased environmental complexity.

- Test 1. Two types of incoming drone, i.e. multi-rotor and fixed-wing, with single maneuverability level.
- Test 2. Two types of incoming drone, i.e. multi-rotor and fixed-wing, with two maneuverability levels.
- Test 3. Two types of incoming drone, i.e. multi-rotor and fixed-wing with a randomly fixed obstacle in the maneuver area.

The observation state includes the ego drone position, the incoming drone position, the ego drone destination, and the incoming drone velocity. The datasets used to test the performance of the proposed method includes 10000 randomly

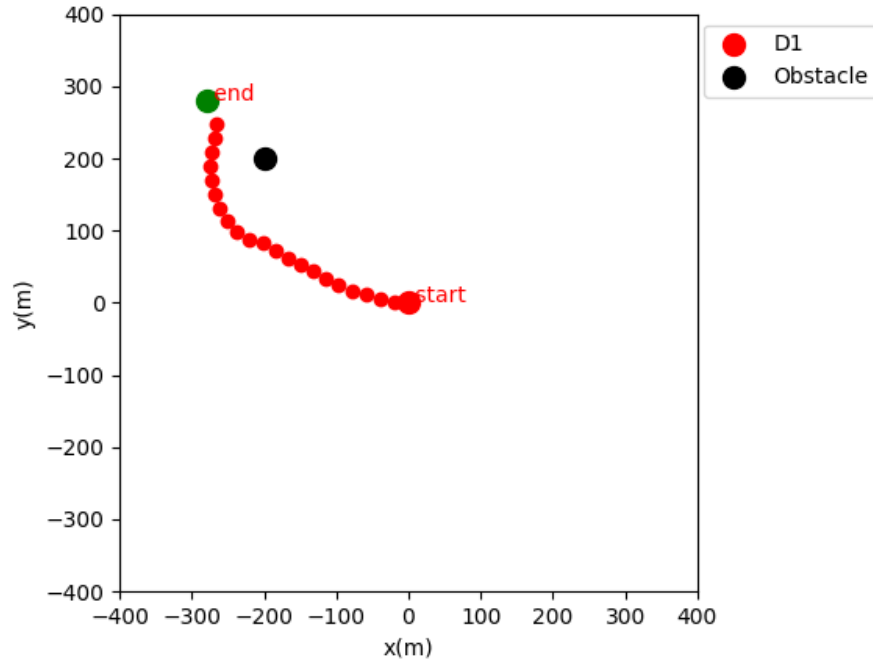


Fig. 5 Examples of SAC algorithm for fixed obstacle avoidance.

selected episodes with a collision scenario according to the involved drones in the bounded area.

A. Test 1

Three models were trained according to the incoming drone type:

- Multi-rotor incoming drone.
- Fixed-wing incoming drone.
- Mixed incoming drone, i.e. randomly selected multi-rotor or fixed-wing drone.

The first model was trained considering a multi-rotor incoming drone and the trajectories generated in Test A and Test B conditions are respectively reported in Fig. 6.

The comparison among multi-rotor, fixed-wing, and mixed models was performed by using datasets with collision events. Table 2 reports the results about collision frequency. The multi-rotor incoming drone case shows better performance mainly due to the smaller velocity value. The model trained with both multi-rotor and fixed-wing incoming drones shows intermediate performance, as expected.

The approach based on single-step maximum distance evaluation allows to solve all collision events for multi-rotor incoming drone case, but it shows worse performance for fixed-wing incoming drone case presenting a frequency of minimum separation violation of 6.66% because the high velocity value and the selected threshold do not allow to solve the collision event in the presented single-step based approach.

Table 2 Performance assessment for Test 1

Incoming drone type	f_{SMI}
MR	0.2%
FW	0.8%
Mixed	0.5%

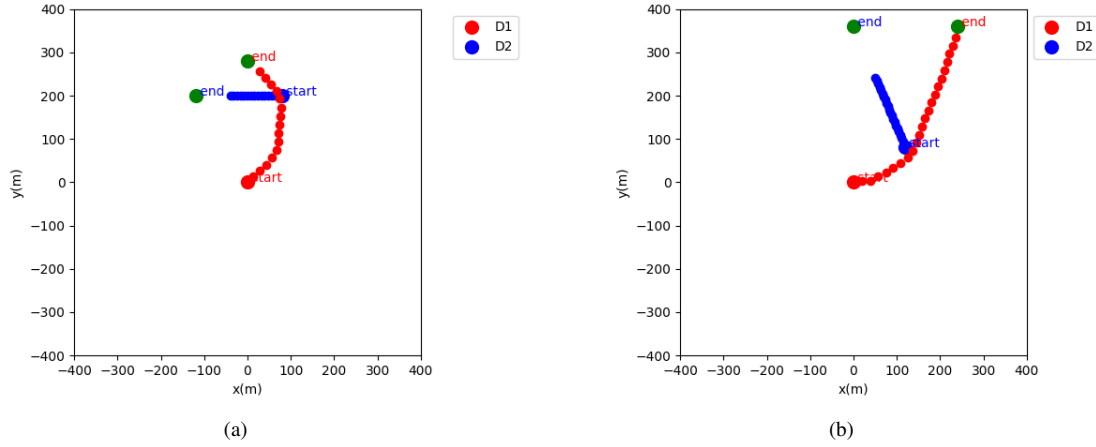


Fig. 6 Examples of proposed method application with a multi-rotor incoming drone for Test 1. a) Test A scenario. b) Test B scenario.

B. Test 2

Different incoming drones were compared to assess the method performance:

- 1) Low maneuverability multi-rotor incoming drone with a constant velocity of 8m/s and a course angle change of 20°.
- 2) High maneuverability incoming fixed-wing drone with a constant velocity of 32m/s and a turning rate of 10°/s.

Thanks to the comparison with Table 2, the results shows that for drones with lower velocity values, the performance has similar results for less aggressive and more aggressive maneuverability levels. Increasing the velocity value, the maneuverability level stronger impacts the results, as reported in Table 3.

Table 3 Performance assessment for Test 2

Incoming drone type	f_{SMI}
Less maneuverability MR	0.2%
High maneuverability FW	1.3%

C. Test 3

A more complex scenario is considered to generate proper maneuvers in order to avoid the incoming drone and the randomly selected fixed obstacle during the same episode. The incoming drone characteristics are the same described in Test 1 scenario and a random fixed obstacle is considered within a circular area of radius 200m around the ego drone starting point. Test A and Test B results are reported in Fig. 7, where a fixed obstacle is added. The results over the same datasets of Test 1 with randomly selected obstacles are reported in Table 4

Table 4 Performance assessment for Test 3

Incoming drone type	f_{SMI}
MR	0.9%
FW	1.9%

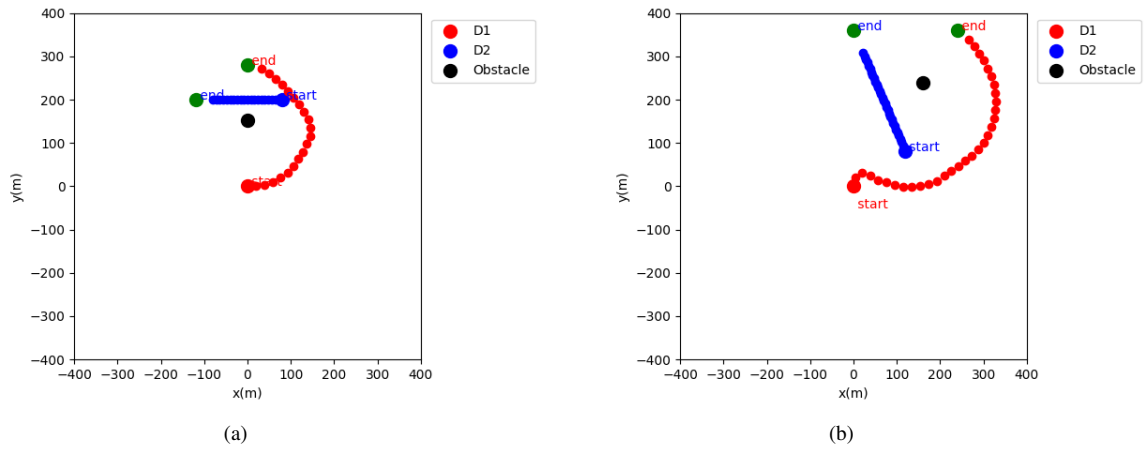


Fig. 7 Examples of proposed method application with a multi-rotor incoming drone and a fixed obstacle. a) Test A scenario. b) Test B scenario.

V. Test Validation

Besides the proposed benchmark, the presented method performance was validated considering different approaches. The first comparison is related to the inclusion of historical position data in the state vector and the second comparison is related to the removal of the drone type information hypothesis.

A. Comparison with Historical Data based Approach

In order to validate the proposed method, historical data related to the incoming drone position are included in the state vector. Specifically, five previous positions are considered removing the information of the incoming drone velocity vector. Fig. 8 shows drones trajectories for Test A and Test B cases. The results over the same datasets of Test 1 are reported in Table 5.

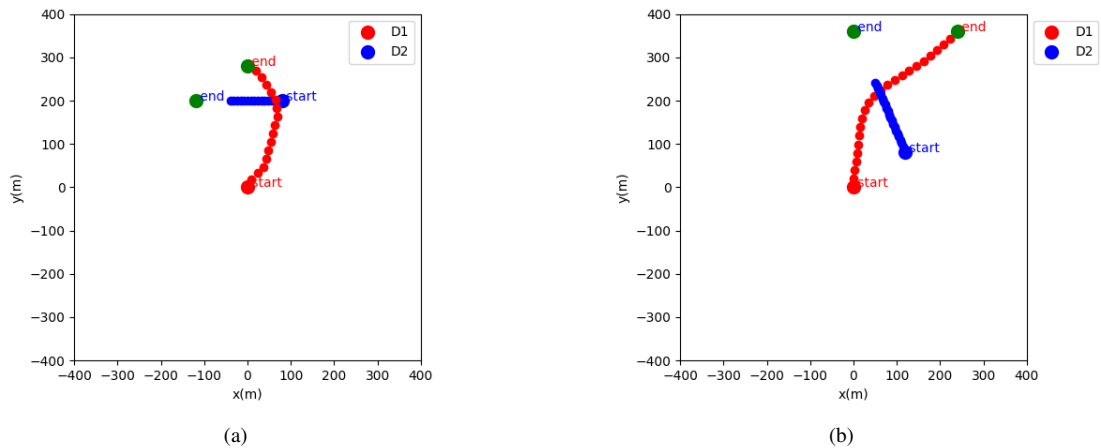


Fig. 8 Examples of the Historical data based Method with a multi-rotor incoming drone. a) Test A scenario. b) Test B scenario.

The results show that the use of velocity vector improves the performance of the presented approach.

Table 5 Performance assessment for Historical Data based Method.

Incoming drone type	f_{SMI}
MR	0.5%
FW	1.8%

B. Comparison with Drone Type Data Removal based Approach

In order to validate the proposed method, the hypothesis related to the previous knowledge about the drone type was removed. Specifically, the method performance was evaluated in case of preliminary misclassification of drone type, i.e. using the model trained with a selected incoming drone for an operative scenario where a different drone type is present. The results over the same datasets of Test 1 are reported in Table 6.

Table 6 Performance assessment for Drone Type Data Removal based Approach

Incoming drone type	f_{SMI} MR dataset	f_{SMI} FW dataset
MR model	0.2%	19.4%
FW model	2.5%	0.8%
Mixed model	0.3%	1.0%

The achieved performance shows that a two-drones-environment with no obstacles results in a Separation Minima Infringement frequency of 0.2% when we have homogeneous agents which increases to 0.8% when we have heterogeneous agents with less maneuverability level. The introduction of a high maneuverability drone and a fixed obstacle involves an increase of the Separation Minima Infringement frequency respectively of 1.3% and 1.9% in the worst case, i.e. with heterogeneous agents. The use of mixed models can help to reduce the risk related to misclassification issues.

VI. Conclusion

The conducted activity is focused on improving the performance of drone collision avoidance when dealing with heterogeneous incoming drones by adding a separate learned module to use a drone-type conditioned learned policy. This paper aims at presenting a Reinforcement Learning based method for evasive maneuver generation in agreement with the incoming drone behavior by exploiting only kinematic information. Several simulations were performed to compare the method performance with different intruder types and obstacles. The achieved results show a Separation Minima Infringement frequency lower than 1% for a single low maneuverability incoming drone scenario and of about 2% for an heterogeneous incoming drone and fixed obstacle scenario, supporting evasive maneuvers in a context of cooperative drones, such as current UTM/UAM scenario.

Further implementations of this approach can extend the simulation environment to include more than two interacting drones in a typical 3D UAM scenario, supporting the implementation of RL techniques in semi-cooperative UAM environments populated by drones with heterogeneous preferences.

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